

COL1000: Introduction to Programming

Lecture 24

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The `import` statement

Standard Import

```
import math
```

Access with: `math.sqrt(16)`

From Import

```
from math import  
sqrt
```

Access with: `sqrt(16)`

Import As

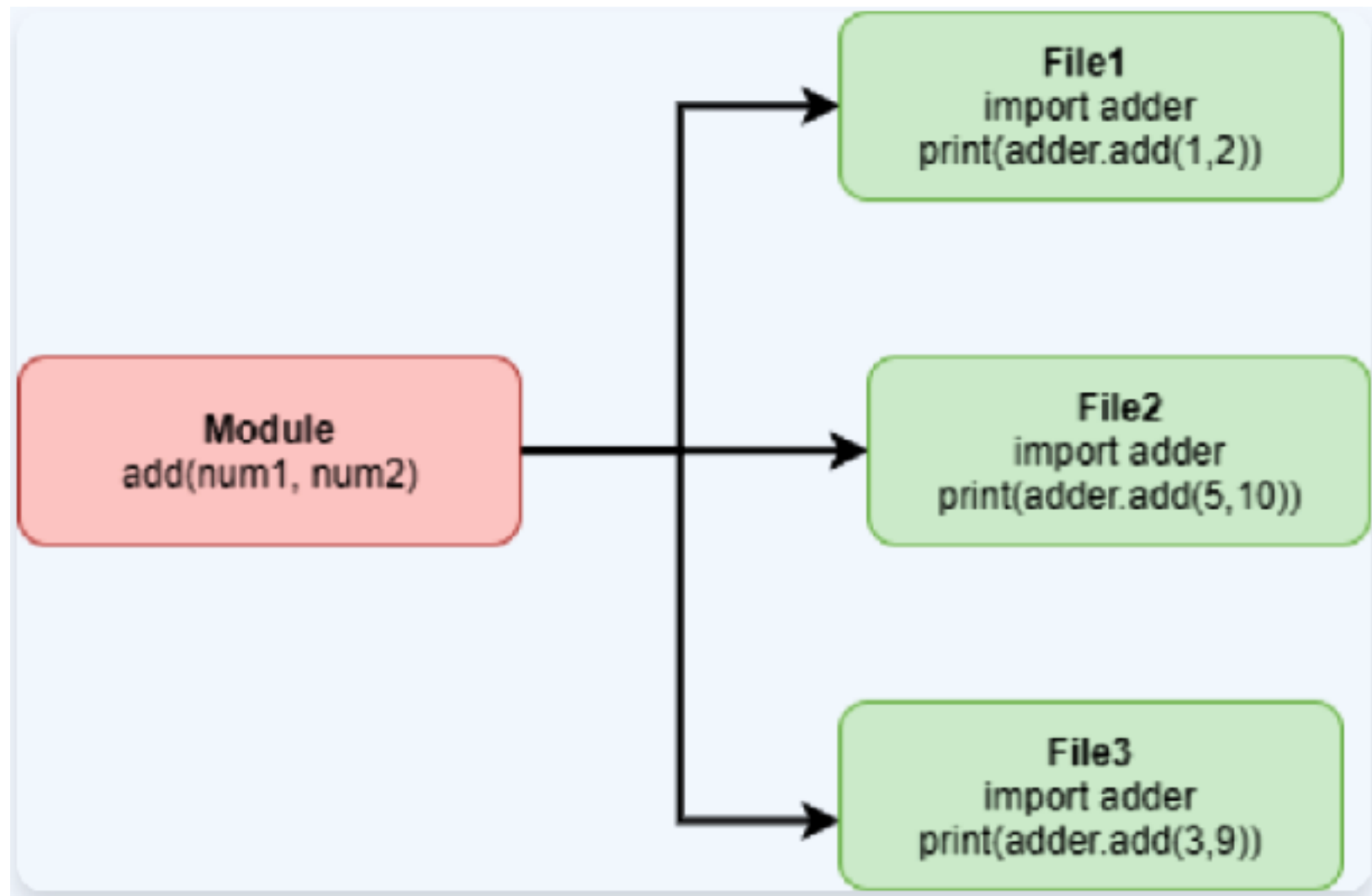
```
import math as m
```

Access with: `m.sqrt(16)`

Caution: If you define your own `sqrt`, then that function will be used, not `math.sqrt`

Modules

- A module is a **.py file** containing Python code
- Helps organise code into separate files
- Makes code reusable across programs
- Allows multiple people to work on different parts



How to use modules

```
# circle.py
```

```
pi = 3.14159
```

```
def area(radius):  
    return pi * radius * radius
```

```
def circumference(radius):  
    return 2 * pi * radius
```

```
# module_usage.py
```

```
import circle
```

```
print(circle.pi) # 3.14159
```

```
print(circle.area(3)) # 28.27431
```

```
import sys
```

```
sys.path.append('/path/to/module_folder')
```

```
import your_module
```

Working with Files

reading input from files

writing output to files

Working with files

'r'

**Read mode
(default)**

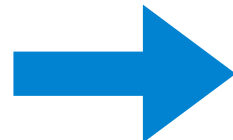
'w'

**Write mode
(overwrites)**

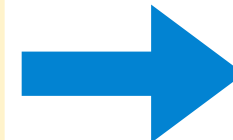
'a'

**Append mode
(writes at end of file)**

**Open the file:
`open('file_path', 'mode')`**



**Perform required operations:
Read / Write / Append**



**Close the file:
`close()`**

Example: Writing to a file

```
handle = open('helloworld.txt', 'w')

# Write data to the file
handle.write('Line 1: Hello World\n')
handle.write('Line 2: Next line.')
handle.write('Still in Line 2.')
handle.close()
```

helloworld.txt

Line 1: Hello World
Line 2: Next line.Still in Line 2.

Writing Process

- Open file with mode 'w' (write)
- Use write() method to add content
- Add '\n' for new lines
- Close the file when finished

Read / Readline

helloworld.txt

Line 1

Line 2

Line 3

Program

```
handle = open('helloworld.txt', 'r')  
  
s = handle.read()  
# reads entire file as one string  
  
print(s)  
handle.close()
```

Output

```
Line 1  
Line 2  
Line 3
```

```
handle = open('helloworld.txt', 'r')  
  
s = handle.readline()  
# reads the current line as string  
  
print(s)  
handle.close()
```

```
Line 1
```


Readline / Readlines

```
# helloworld.txt
```

```
Line 1
```

```
Line 2
```

```
Line 3
```

Program

```
handle = open('helloworld.txt', 'r')
s = handle.readline()
t = handle.readlines()
# list of strings from current line

print(t)
handle.close()
```

Output

```
['Line 2\n', 'Line 3']
```

```
handle = open('helloworld.txt', 'r')
s = handle.readline()
handle.close()

handle = open('helloworld.txt', 'r')
t = handle.readlines()
print(t)
handle.close()
```

```
['Line 1\n',
 'Line 2\n', 'Line 3']
```

Reading lines using loops

helloworld.txt

Line 1
Line 2
Line 3

Program

```
handle = open('helloworld.txt', 'r')  
  
s = '...'  
  
# loop reads one line at a time  
while s != '':  
    s = handle.readline()  
    print(s)  
  
handle.close()
```

Output

```
Line 1  
  
Line 2  
  
Line 3  
—
```

```
handle = open('helloworld.txt', 'r')  
  
# loop reads one line at a time  
for s in handle:  
    print(s)  
  
handle.close()
```

```
Line 1  
  
Line 2  
  
Line 3  
—
```

Opening file using “with”

```
# helloworld.txt
```

```
Line 1  
Line 2  
Line 3
```

Program

```
with open('helloworld.txt', 'r') as handle:  
    print(handle.readline())  
    print(handle.readlines())  
  
# no need to close if read/write using with
```

Output

```
Line 1  
['Line 2\n', 'Line 3']  
_
```

```
with open('helloworld.txt', 'r') as handle:  
    print(handle.readline())  
  
print(handle.readlines())
```

```
Line 1  
  
ValueError: I/O operation  
on closed file
```

Exercise: storing primes in files

Input: number $n > 3$

Produce five files:

ending_1.txt:

ending_3.txt

ending_5.txt

ending_7.txt,

ending_9.txt

where ending_i.txt stores
all primes between 3 and n
that end with the digit i.

```
def check_prime(x):  
    if(x==1): return False  
  
    for i in range(2, x):  
        if(x%i == 0): return False  
        if(i*i > x): break  
  
    return True  
  
n = int(input("Enter n: "))  
  
for j in range(3, n+1):  
    if(check_prime(j)):  
        last = str(j%10)  
  
        f = "ending_"+last+".txt"  
  
        with open(f, 'a') as handle:  
            handle.write(str(j)+"\n")
```